

Cuba’s depopulation, 2019–2026: Monte Carlo interval estimates, an infant-mortality sentinel, and triangulation of independent registers

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Working paper (preprint) · 30 July 2026 · Not peer-reviewed

Abstract

Background. Cuba is losing population at one of the fastest peacetime rates in the Western Hemisphere. No census has been completed since 2012, and the national statistics office (ONEI) removed more than one million people from its published headcount in a single 2024 revision. End-2024 estimates still span roughly 8.0–11.0 million. This paper quantifies the 2019–2025 decline, decomposes migration versus internal demographic collapse, and projects to end-2026.

Methods. We solve the demographic accounting identity with Monte Carlo simulation ($N = 10^6$) under four prior sets (Models A–D). Preferred Model D anchors mortality in infant mortality—a high-integrity yearly sentinel—via an externally calibrated elasticity, and is validated by triangulation across independent administrative sources. Age-standardized excess mortality (Kitagawa-style) is reported separately from crude registered excess.

Results. Preferred Model D places the living population near **8.56 million** at end-2025 (median loss **2.31 million, 21.3%** on a corrected 2019 base; 90% CI for loss roughly 1.94–2.71 million). Emigration accounts for about **86%** of the absolute loss. Age-adjusted excess deaths in 2024–2025 are about **40,000–60,000**; crude registered excess over 2020–2025 is larger ($\sim 153,000$) and must not be conflated with the age-adjusted figure. A nowcast places Model D near **8.27 million** by end-2026 ($\sim 290,000$ further loss).

Conclusions. Depopulation is predominantly migratory in the headcount, but is reinforced by collapsing births, elevated mortality, and accelerated ageing. Cumulative loss from 2019 approaches roughly one in four people by end-2026 when measured against the official 2019 base—a different definition from the Model D -21% figure for end-2025.

Keywords: Cuba; demography; emigration; natural decrease; infant mortality; fertility; ageing; statistical data quality.

1 Introduction

Counting Cubans has become contested. The last completed census was in 2012; the census planned for 2022 has been repeatedly postponed [ONEI, 2022]. In July 2024, ONEI reported an “effective population” of 10,055,968 at end-2023—about one million below prior published levels—then 9,748,007 (end-2024) and 9,434,593 (end-2025) [ONEI, 2024, 2025].

Three incompatible series coexist (Figure 1): the United Nations World Population Prospects path near 10.9 million (still assuming net emigration of order 10^4 per year); ONEI’s revised official series (9.43 million at end-2025); and independent estimates placing the living population nearer 8.0–8.6 million [Albizu-

Campos Espiñeira, 2023, 2024]. This paper estimates the 2019–2025 decline with honest intervals, decomposes drivers, audits official migration accounting, and projects to end-2026.

Central thesis. The headcount loss is mostly emigration, but a second “internal” engine—birth collapse, excess mortality, and ageing—would sustain contraction even if emigration slowed.

2 Methods

2.1 Accounting identity and Monte Carlo

Population change obeys

$$P_{\text{end}} = P_{\text{start}} + B - D - M_{\text{net}}, \quad (1)$$

where B , D , and M_{net} are births, deaths, and net emigration over 2020–2025. We sample $N = 10^6$ internally consistent scenarios from bounded priors, with correlation induced by a Gaussian copula. Within-model 90% intervals are statistical uncertainty; the **range across models** (~ 1.8 – 3.0 million of cumulative loss) is structural uncertainty and should not be read as a single CI [Pérez-Riverol, 2026c].

2.2 Four models

- **A (conservative):** ONEI-anchored vital events and migration.
- **B (crisis-adjusted):** limited death under-registration plus higher migration toward independent estimates.
- **C (worst case):** upper-bound priors calibrated to peacetime collapse analogs.
- **D (preferred, sentinel):** death under-registration tightly bounded using infant mortality as a sentinel; migration uncertainty retained as in B/C.

2.3 Infant-mortality sentinel

Infant mortality rose from 5.0 to 9.9 per thousand (2019–2025), a $\sim 98\%$ increase (and $\sim 39\%$ year-on-year in 2024–2025). Using an elasticity $e \approx 0.30$ from collapse analogs (range 0.22–0.40), the predicted all-cause mortality rise matches the registered crude death-rate rise ($\sim 33\%$ by 2024), arguing against huge hidden institutional deaths and locating residual under-registration mainly in at-home elderly deaths during funeral-system stress [Pérez-Riverol, 2026b].

2.4 Excess mortality

We distinguish (i) **crude** registered excess versus a no-crisis counterfactual on crisis-adjusted mid-year stocks ($\sim 153,000$ over 2020–2025; $\sim 66,000$ in 2024–2025), from (ii) **age-standardized** excess answering “deaths beyond ageing”—about **40,000–60,000** in 2024–2025 (primary figure for science and communication).

2.5 Triangulation

Eight independent levels (UN, MINSAP denominator, electoral roll, ONEI, housing occupancy, Albizu-Campos 2023/2024, and Model D) place the probable band near 8.0–8.9 million; registers that do not purge emigrants act as ceilings (Figure 3).

2.6 Destinations ledger

Settled migrants only (avoiding transit double-counting): United States ~800,000; Spain ~130–135,000; Uruguay ~35,000; other ~255,000; floor ~1.05 million. Encounter counts and nationality applications are separate concepts and are not added into the settled floor [Pérez-Riverol, 2026a].

3 Results

3.1 Headline estimates

Table 1 summarizes end-2025 medians. Model D yields median population **8.56 million** and loss **2.31 million** (21.3% on the corrected base). Migration's share of absolute loss is about **86–88%**. A 2026 nowcast under continued crisis conditions places Model D near **8.27 million** (further loss ~290,000).

Table 1: Four models: population loss 2019–2025 and end-2025 population (medians).

Model	Main assumption	Loss (M)	%	Pop. end-2025 (M)
A · conservative	ONEI-anchored	1.93	17.6	9.06
B · crisis-adjusted	under-reg. + independent migration	2.31	21.3	8.54
C · worst case	analog upper bound	2.53	23.5	8.24
D · sentinel (preferred)	IMR sentinel + triangulation	2.31	21.3	8.56

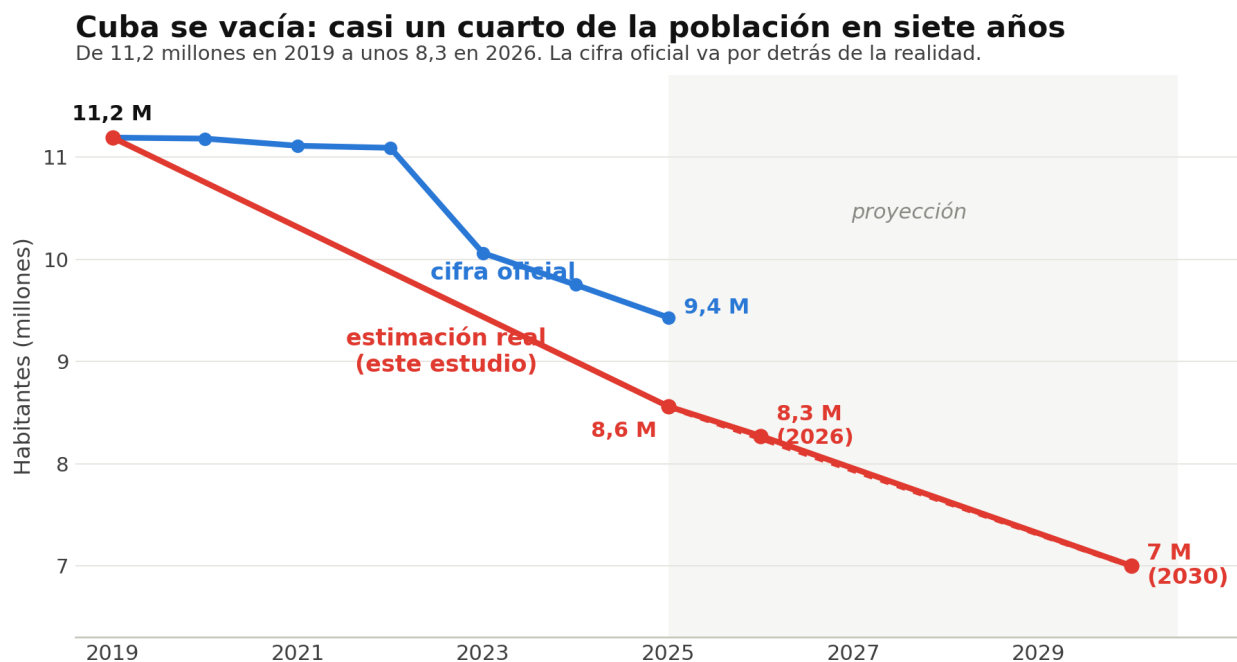


Figure 1: Official versus estimated living population, 2019–2030 (Model D path in red).

3.2 Internal engine

Births fell to 68,064 in 2025 (TFR 1.29); deaths were 136,214; natural balance –68,149. About 77% of emigrants are aged 15–59; among those remaining, about 26.7% are 60+ (median age ~45).

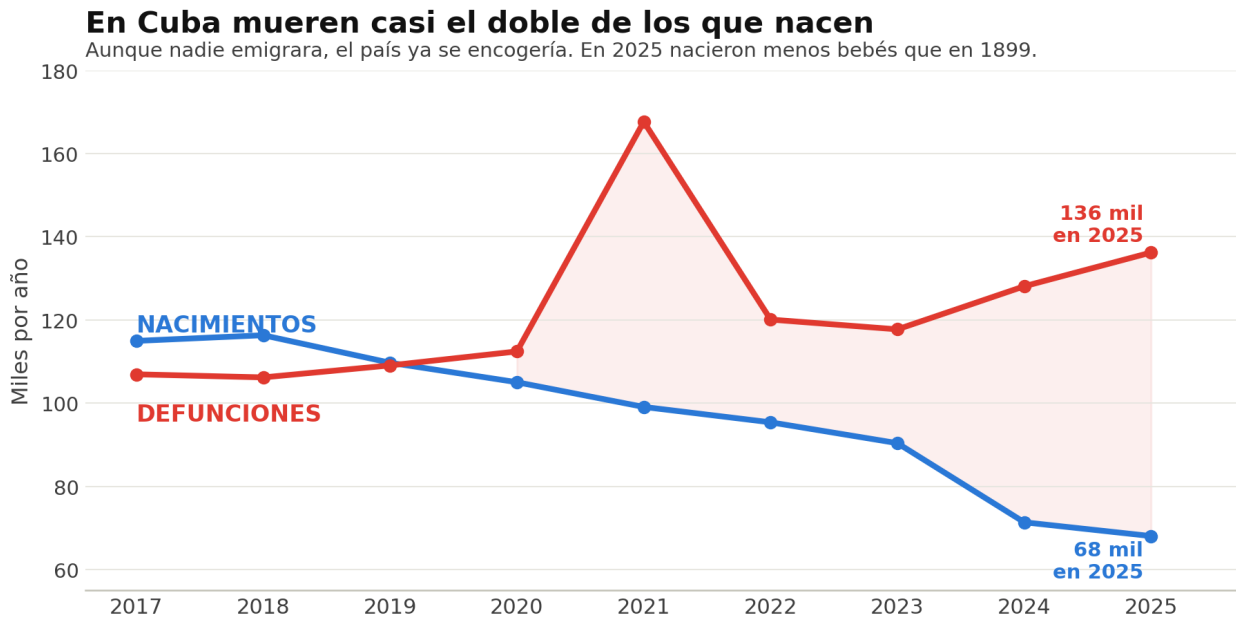


Figure 2: Births versus deaths (“scissors”): deaths roughly double births by 2025.

4 Discussion

Model D is preferred because the sentinel validates that registered deaths move with crisis severity, shrinking the need for large hidden-death multipliers and focusing residual uncertainty on migration and triangulation. An adversarial review correctly stresses that the loss is $\sim 86\%$ migratory and that within-model intervals can look falsely precise relative to across-model structure; those caveats are retained.

Percent language. Against the official 2019 base (11.19 M), cumulative loss by end-2026 approaches $\sim 25\%$ (“about one in four”). Against the corrected Model D base, end-2025 loss is $\sim 21\%$ (“about one in five”). These are not interchangeable headlines.

5 Limitations

No post-2012 census; ONEI’s ~ 24 -month emigration rule lags recent exits; destination microdata are incomplete; elasticity and housing-occupancy assumptions are approximate; 2030 paths are scenarios, not forecasts.

6 Data and code availability

ONEI tables and a machine-readable claim sheet are in the public repository <https://github.com/ypriverol/cubascience> under `demographics/`. Interactive companion: <https://ypriverol.github.io/cubascience/demographics/>. Monte Carlo scripts (`scripts/model_*.py`) write summaries to `artifacts/`. Third-party literature is cited, not redistributed.

Competing interests

None declared.

Cuántos cubanos quedan: seis formas de estimarlo

Las fuentes que no borran a los que se fueron dan cifras altas; las que sí lo corrigen convergen en ~8,5 millones.

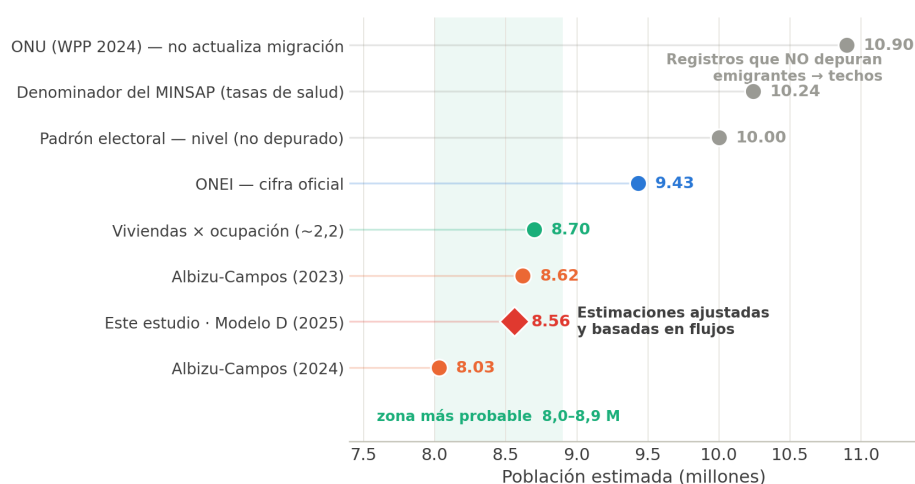


Figure 3: Triangulation of end-2025 population levels across independent sources.

Aun descontando el envejecimiento, mueren más de lo normal

En 2024-2025 hubo ~40 000-60 000 muertes por encima de lo esperado; las tasas de los mayores de 60 están ~11-27% sobre 2019.

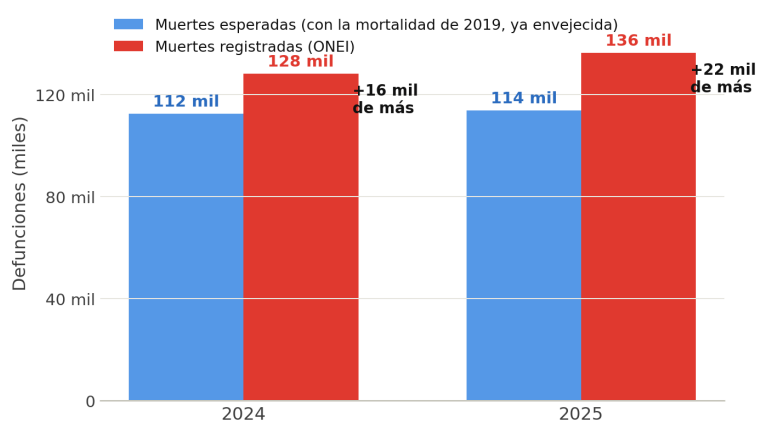


Figure 4: Age-context for excess mortality; primary communication figure is 40–60 thousand age-adjusted excess deaths in 2024–2025.

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